

Sample sizes in sport and exercise psychology: Have they increased in the past decade?

Tadeus Pindl^{a,b}, Geoffrey Schweizer^{a,*}, Philip Furley^c

^a Heidelberg University, Department of Sport and Exercise Psychology, Im Neuenheimer Feld 720, Heidelberg, 69120, Germany

^b Heidelberg University, Department of Quantitative Research Methods, Institute of Psychology, Hauptstraße 47 - 51, Heidelberg, 69120, Germany

^c German Sport University, Cologne, Institute of Training and Computer Science in Sports, Am Sportpark Müngersdorf 6, Köln, 50933, Germany

ARTICLE INFO

Keywords:

Effect size
Meta science
Power
Replicability
Research methods

ABSTRACT

Although the past decade was characterized by methodological discussions regarding the quality of psychological research, there is sparse empirical evidence whether research practices have changed since the beginning of the debate. We aim to investigate whether sample sizes in sport and exercise psychology have increased in the past decade.

We analyze four leading journals in sport and exercise psychology regarding sample sizes of all quantitative studies published in these journals between 2022 and 2023. Then we compare these sample sizes to the sample sizes published in the same journals between 2009 and 2013 (Schweizer & Furley, 2016).

Our results suggest that sample sizes in sport and exercise psychology have increased for most, but not all study designs (e.g., median sample sizes increased by 44% and 33% for experimental and correlational studies, whereas they decreased by 25% for quasi-experimental studies). Although power may have increased accordingly, our results suggest that a substantial proportion of studies in sport and exercise psychology may still not have sufficient power to detect small-to-medium effects, particularly so when testing at low alpha levels. More than half of the studies in our sample do not report an a priori power analysis, and a priori power analyses are not uniformly related to sample sizes across study designs.

Taken together, the present study suggests that sample sizes in sport and exercise psychology have increased in the past decade. However, there is still room for improvement. In order to remedy this shortcoming, we suggest investing in training how to perform a priori power analyses and in creating applications that make it easier to conduct respective power analyses.

"If a team of research psychologists were to emerge today from a 7-year hibernation, they would not recognize their field. Authors voluntarily posting their data. ... Hundreds of researchers preregistering their studies. ... Enormous increases in sample sizes" (Nelson et al., 2018, p. 512).

Starting in the early 2010s, the quality of psychological research has been the subject of an intense and highly critical debate (Hardwicke et al., 2020; Nelson et al., 2018). As a consequence, a multitude of measures have been proposed and implemented to improve the quality of psychological research (for overviews see Hardwicke et al., 2020; Munafò et al., 2017; Nelson et al., 2018). These measures include, but are not limited to, preregistering analysis plans in order to prevent p-hacking (Hardwicke & Wagenmakers, 2023), providing data, code and material in order to make analyses reproducible (Nosek, 2015), and

guidelines for authors and reviewers in order to make analysis choices transparent (Nosek et al., 2015; Simmons et al., 2011). Furthermore, researchers are called to ensure the reliability and validity of their measurements (e.g., Schimmack, 2021; Schweizer et al., 2020; Westfall & Yarkoni, 2016), to justify their sample sizes (Lakens, 2022) and to plan them more adequately (e.g., Fraley & Vazire, 2014; Sommet et al., 2023), to lower their alpha levels (Benjamin et al., 2018) and to employ more sophisticated statistical procedures (e.g., Lakens et al., 2018; Lakens & Evers, 2014; Wagenmakers et al., 2018).

Despite literally hundreds of methodological articles, the creation of new journals and scientific societies, the success of blogs dedicated to methodological criticism, the development of web tools and the adoption of requirements for authors and reviewers, it seems hard to say whether psychological research has really changed for the better as

* Corresponding author.

E-mail address: geoffrey.schweizer@issw.uni-heidelberg.de (G. Schweizer).

<https://doi.org/10.1016/j.psychsport.2026.103231>

Received 16 January 2026; Received in revised form 29 April 2026; Accepted 20 July 2026

Available online 21 July 2026

1469-0292/© 2026 Published by Elsevier Ltd.

claimed by Nelson et al. in the quote at the beginning of this article (2018). Some authors adopt a very optimistic perspective. For example, Nelson et al. (2018, p. 512) go so far as to proclaim “psychology’s renaissance” and assert that “the improvements to our field have been dramatic”. Other authors remain more sceptic. For example, Hardwicke et al. report evidence for a “... minimal adoption of transparency and reproducibility-related research practices” (2022, p. 247).

Similarly, in the field of sport and exercise sciences, several recent publications have highlighted methodological challenges parallel to the ones in psychology (Mesquida et al., 2022; Murphy et al., 2025; Tiller & Ekkekakis, 2023; Uehata & Rein, 2025). Contrary to the optimistic outlook in psychology (e.g., Nelson et al., 2018), however, Tiller and Ekkekakis (2023, p. 201) have suggested that researchers in the field of sport and exercise sciences might exhibit “a reluctance to acknowledge their methodological shortcomings” and thus hesitate to improve methodological practices. However, again there is little evidence to support this claim. Thus, both in the fields of psychology and of sport and exercise sciences, it is hard to say to what extent research practices have improved in the wake of the current methodological discussion. The rather pessimistic outlook expressed by Tiller and Ekkekakis (2023) may reflect the fact that the discussion around the quality of research practices has not yet reached the same extent in the field of sport and exercise sciences as it has in psychology. At the same time, one reason for this difference might be that at least in some fields of sports sciences (e.g., physiology), effects may be larger and measurement may be more precise, thus necessitating less critical inquiry.

One major reason for the ambiguity described above is the lack of pre-post data that could be used to quantify developments or improvements in psychological or sport scientific research (see Hardwicke et al., 2022, for a similar reasoning). In other words, objectively assessing improvements in psychological or sport scientific research would require the existence of a metric that a) rather unambiguously captures a characteristic of high-quality research, b) had room for improvement at the beginning of the discussion on research quality and c) is available for studies before and after the onset of the respective discussion. Several of such metrics exist, one of them being sample sizes. Compared to other metrics, one advantage of sample sizes is that they are easy to assess, since they are directly reported in every research paper. Therefore, the goal of the present research is to investigate whether research practices in sport and exercise psychology have changed since the beginning of the current methodological discussion, using comparisons of sample sizes as an indicator.

1. The role of sample sizes for research quality

Within the above-mentioned discussion on the quality of research practices in psychology, a particular emphasis has been on studies’ power (Brysbaert, 2019; Button et al., 2013; Fraley & Vazire, 2014). Power is defined as the probability of detecting an effect, given that said effect exists (Button et al., 2013), or, as the probability of correctly rejecting a null hypothesis (Button et al., 2013). Several studies suggest that psychological research is probably underpowered, because sample sizes are smaller than they need to be for obtaining adequate levels of power (Button et al., 2013; Fraley & Vazire, 2014; Sommet et al., 2023). Compared to fields with lower power, fields with higher power have on average a lower proportion of false-positive and a lower proportion of false-negative results (Button et al., 2013; Wilson & Wixted, 2018). In other words, when power increases, researchers have a higher probability of finding an effect given that it exists and a lower probability of reporting an effect that in reality does not exist (Button et al., 2013; Wilson & Wixted, 2018). Accordingly, the average power of studies in a field influences the level of trust that researchers, practitioners, and the general public alike should place in this very field’s results.

In 2014, Fraley and Vazire estimated the average power of studies in social and personality psychology. According to their estimates, a large proportion of studies in these fields could be considered substantially

underpowered, at least in order to detect the small-to-medium effect sizes that are considered typical in social and personality psychology. However, studies’ power would have been more adequate in order to detect larger effect sizes.

Furthermore, Fraley and Vazire suggested adopting a new metric for evaluating journals based on the average sample size of studies in a journal. Drawing on the term ‘impact factor’, they called this metric the N-Pact factor. Inspired by the work of Fraley and Vazire, analyses of sample sizes and respective estimates of studies’ power followed in several fields (Kossmeier et al., 2019; Reardon et al., 2019; Schweizer & Furley, 2016). In 2016, Schweizer and Furley analyzed the sample sizes of 720 studies published between 2009 and 2013 in four major journals from sport and exercise psychology. Furthermore, they expanded upon the analyses by Fraley and Vazire (2014) by analyzing sample sizes separately for studies’ design. Results suggest that sample sizes in sport and exercise psychology heavily depend on studies’ design, and that, therefore, differences between journals are diminished when controlling for design. The results by Schweizer and Furley suggest that the median sample sizes of correlational studies and mixed-design experiments in the respective period were adequate in order to detect an effect size of $r = .2$ (which might be considered the typical effect size in psychological research; Richard et al., 2003; see Sommet et al. [2023] for a more recent summary of the discussion of effect sizes in psychological research), however, that the median sample sizes of quasi-experimental studies, within-participants experiments and between-participants experiments had less than a 50% chance to detect an effect of $r = .2$. Furthermore, variations regarding sample sizes were substantial, suggesting that while 50% of studies had more power than reported above, 50% of studies also had less power. Thus, even for study categories where the median sample size may have resulted in adequate power in order to detect small-to-medium effect sizes, there were probably a substantial proportion of studies (namely those below the median) where it did not.

It seems important to discuss whether effect sizes in sport and exercise psychology might be larger than published estimates of average effect sizes in psychological research suggest (see Sommet et al., 2023, for a discussion and brief summary of average effect sizes in psychological research). On the one hand, one might argue that most published estimates stem from rather theoretical fields (e.g., social or personality psychology), where researchers may be interested in comparably small effects. In more applied fields (such as sport and exercise psychology), average effect sizes might be larger, simply because researchers are not interested in effect sizes that are too small to be meaningfully applied. Indeed, recent data suggest that published effect sizes from applied psychology might be larger than published effect sizes from other psychological fields (Schäfer & Schwarz, 2019). On the other hand, all estimates of average effect sizes are based on *published* effect sizes, and there are reasons to believe that true effects are often smaller than published effect sizes. First, when sample sizes are small, only large effects become significant and get published (Button et al., 2013). Second, effect sizes found in replication studies are often smaller than effect sizes found in original studies; and effect sizes from preregistered studies are often smaller than effect sizes from non-preregistered ones (see Sommet et al., 2023, for a brief overview and further references). Furthermore, some interventions will have small effects (e.g., Milkman et al., 2021). In line with these observations, recent data suggest that even in a highly applied field such as increasing the frequency of exercising, researchers and practitioners alike overestimate the efficacy of interventions (Milkman et al., 2021). Taken together, it seems important to keep in mind that particularly in applied fields, researchers may be interested in rather large effects, and that for these researchers, smaller sample sizes will not constitute underpowered research (still, effect size estimates will be less precise than with larger samples). At the same time, effect sizes may often be smaller than researchers anticipate.

Additionally, sport science in general and sport psychology in particular may differ from social, cognitive and personality psychology

(where the discussions around sample sizes and research quality originated) with regard to resource constraints. Obtaining large samples may be more costly in sport science or sport psychology, because researchers may be interested in specific populations (e.g., experts, high-performance athletes), that are more expensive to sample than the student or MTurk participants common to some fields of psychology. Furthermore, measurement itself may be more costly, as sports scientific or sport psychological studies may rely on behavioral measurements instead of online questionnaires or reaction-time tasks (e.g., see Schweizer et al., 2020, for the construction of reliable performance outcome measures in sport psychology). On the other hand, research in sport science or sport psychology may sometimes sample entire populations, or at least nearly entire populations (e.g., all top athletes in a field). When sampling (nearly) the whole population, there is no need for significance tests or larger samples (Lakens, 2022), at least when the goal is to make an inference regarding the respective population.

Following the power estimates reported above, Schweizer and Furley (2016) called for more attention to the issue of sample sizes and power in the field of sport and exercise psychology. Consequentially, they suggested that researchers should calculate their sample sizes based on anticipated effect sizes. If researchers followed this advice, future sample sizes should increase, considering that a substantial proportion of studies in sport and exercise psychology was probably underpowered, at least in order to detect small-to-medium effect sizes. The publication did not only lead to calls and suggestions, but it also had consequences on editorial policy. For example, the journal 'Psychology of Sport and Exercise' updated their author requirements regarding sample sizes accordingly.

The goal of the present paper is to investigate whether sample sizes – as one instance of methodological practices – in sport and exercise psychology have increased since the beginning of the current methodological discussion. In order to answer this question, we aim to use the data provided by Schweizer and Furley (2016) as a benchmark.

2. Method

In order to ensure comparability with the data from 2009 to 2013, we closely followed the approach described in Schweizer and Furley (2016). We examined studies from the same four journals in sport and exercise psychology, namely International Journal of Sport Psychology (IJSP), Journal of Applied Sport Psychology (JASP), Journal of Sport and Exercise Psychology (JSEP), and Psychology of Sport and Exercise (PSE). We examined all empirical articles published in these journals in the years 2022 and 2023 (due to resource constraints we could not replicate a full five-year period). We did not examine non-empirical articles such as review articles and comments. In a first step, we identified empirical articles. In the next step, among empirical articles, we did not analyze meta-analyses, studies based on the analysis of data bases, and qualitative studies such as interview studies or focus groups. Other reasons for not including empirical studies were descriptive approaches, studies in which we could not identify the relevant sample size, and studies with analyses that do not necessarily rely on large samples (e.g., single case studies). Furthermore, we excluded studies that served the purpose of constructing a questionnaire.

When an article contained multiple studies that were analyzed separately, we coded each study separately. For every study, we coded in which journal it was published, publication year, sample size, and whether it was an experimental, a quasi-experimental, or a correlational study. We coded a study as experimental, when there was a randomized assignment of participants to factors (between-participants design) or a repeated manipulation (within-participants design). Furthermore, we distinguished between within-participants, mixed (a combination of a within-participants and a between-participants factor), and between-participants designs. We coded studies as quasi-experimental, when participants' assignment to groups was based on an a-priori existing variable instead of being randomized (e.g., studies on differences

between experts and novices). We coded a study as correlational, when there was no quasi-experimental or randomized assignment of participants to factors. Furthermore, studies using a multilevel design and studies using structural equation modeling were also coded as correlational. When coding study design, we focused on the operationalization of the main hypotheses. For example, studies, in which hypotheses referred to an experimental design, however contained some additional correlational analyses, were coded as experimental. For studies using a mixed-methods approach, the sample size for the quantitative part was included if it was used in order to answer a research question, as opposed to only supplementing the qualitative analysis.

We coded the initial sample size before cases were excluded. We made exceptions to this rule when the initial sample was only used in order to screen for and identify subgroups that would subsequently be used for the main analyses. In longitudinal studies, we used the sample size at the first measurement time. In dyadic studies, we used the sample size that was the unit of analysis (i.e., in some dyadic studies this is the dyad, in others it is each individual). Furthermore, and novel to this study, we coded whether an a priori power analysis was reported. We did not code post-hoc power analyses (Lakens, 2014).

One author coded all studies. A second author coded all studies again. When both coders disagreed, they discussed their disagreement and solved it. During the coding process, the coders focused on the comparability of the present coding procedure with the procedure employed for the data from 2009 to 2013. We provide all data and R code for the analyses in an online supplement (<https://doi.org/10.23668/psycharchives.22334>). For descriptive analyses, RStudio version 2023.12.1 + 402 (R version 4.3.3) was used (R Core Team, 2023). In order to ensure comparability with the data from 2009 to 2013, all power analyses were conducted in G*Power 3.1.9.7. (Faul et al., 2007). In the next step, all power analyses were conducted again in R using the 'Webpower' package in order to ensure reproducibility of the present data (Zhang & Mai, 2023). Both methods led to the same results. We present information on the power analyses in an online supplement and in the file containing the R code (please see the link above for more information on the power calculations). We do not present data separately for different journals, for two reasons: First, the results from Schweizer & Furley suggest that differences between journals can mostly be explained by differences in study designs (e.g., some journals report more correlational studies than others). Second, we do not aim to present a new metric for evaluating journals (as Fraley & Vazire [2014] originally intended), but instead we aim to analyze potential changes in the last decade.

3. Results

In total, we analyzed the sample sizes of 326 studies (see Table 1 for information how these studies were distributed across study designs and the investigation period). These 326 studies originate from 300 papers, suggesting that the majority of papers in our sample (92%) were single-study papers. Most studies in the sample were coded as correlational (52%), followed by experimental (37%) and quasi-experimental (11%) studies. When interpreting the following data on sample sizes, it is important to keep in mind that the different study designs were not equally frequent in our sample. Main reasons for not including studies

Table 1
Number of analyzed studies per year and design category.

	2022	2023	Total
Experimental (all)	56	64	120
Experimental (Within)	16	19	35
Experimental (Mixed)	34	40	74
Experimental (Between)	6	5	11
Quasi-Experimental	20	17	37
Correlational	92	77	169
Total	168	158	326

were that they employed a qualitative approach and that they aimed at questionnaire construction (see the data in the online supplement for detailed information on the different reasons for excluding a study).

We present median sample sizes and quartiles (i.e., Q25, Q75) separately for experimental, quasi-experimental and correlational studies. Among experimental studies, we present data separately for within-participants, between-participants and mixed designs. Data suggest that sample sizes increased for correlational studies and all subcategories of experimental studies (i.e., within-participants, between-participants and mixed designs), however they decreased for quasi-experimental studies (see Table 2 for a comparison of sample sizes between 2009–2013 and 2022–2023). Importantly, when sample sizes increased, not only median sample sizes but also quartiles increased, suggesting that the whole distribution of sample sizes was shifted to the right (see Fig. 1 for a visual comparison of sample sizes between 2009–2013 and 2022–2023).

The increases that we find for experimental and correlational studies are substantial, as indicated by the relative changes in percentages (Table 2). For experimental studies, median sample sizes increased by 44% (all experimental studies). When looking at different experimental designs separately, median sample sizes increased by 54% (within-participants studies), 32% (mixed-design studies) and 76% (between-participants studies), respectively. For correlational studies, median sample sizes increased by 33%. At the same time, sample sizes decreased by 25% for quasi-experimental studies.

In order to assess whether these changes are within the range of normal fluctuations, we compared the median sample sizes for 2022–2023 to the fluctuations between 2009 and 2013 (Fig. 2). Fig. 2 shows that both for correlational and experimental studies, the median sample sizes for 2022–2023 are outside the fluctuations between 2009 and 2013, suggesting that the reported increase is a systematic one.¹ For quasi-experimental studies, the median sample size in 2022–2023 is lower than the respective sample sizes in 2010–2013, but higher than in 2009, making the picture less clear.

Furthermore, we calculated statistical power for the median sample sizes from the studies in our sample using G*Power and the ‘WebPower’ package in R (Faul et al., 2007; Zhang & Mai, 2023; see the respective online supplement [https://doi.org/10.23668/psycharchives.22334] for more information on the power calculations). We calculated power a) for a range of typical effect sizes (Richard et al., 2003; Sommet et al., 2023), b) for different significance levels and c) separately for correlational, experimental and quasi-experimental studies. Table 3 suggests that power heavily depends on the respective combinations of designs, significance levels, and effect sizes. When comparing 2009–2013 to 2022–2023, power estimates obviously follow the development of

sample sizes: In study categories where sample sizes have increased, power has increased accordingly (when assuming equal effect sizes and significance levels), and vice versa. Still, the present estimates suggest that although many studies in sport and exercise psychology may achieve adequate power, there is still a sizable portion of studies that will not achieve satisfactory levels of power. This is particularly true for studies at the lower end of the sample-size distribution. Furthermore, most power estimates can only be considered satisfactory for $\alpha = .05$, and not for $\alpha = .01$. Given that employing significance levels of $\alpha = .01$ or even lower seems beneficial for lowering the number of false positives while not affecting false negatives (when power is kept constant) (see Benjamin et al., 2018; Miller & Ulrich, 2016), this indicates room for further improvement.

We want to make clear (as this is a common misunderstanding), that we did not calculate the studies’ post-hoc power for finding the effects that these studies reported to find. Instead, we estimate, based on their average sample size, which power the studies in our sample would have had in order to find effects of a certain size given certain significance levels.

As a complement to Table 3, which depicts studies’ power to detect certain effect sizes as a function of median sample sizes and selected significance levels, we report Table 4. Table 4 depicts medium detectable effect sizes (MDES, Bloom, 1995) as a function of studies’ median sample sizes and selected significance levels. In other words, whereas Table 3 describes which power the studies in our sample would have in order to detect a certain effect size based on the median sample sizes that we found, Table 4 describes which effect sizes the studies in our sample would be able to detect with a certain level of power, again based on the median sample sizes we found. Table 4 suggests, that, for many levels of power and choices of significance levels, effect sizes must be large in order to be detectable based on the sample sizes we found in our sample of studies. Importantly, this is not only true for potentially extreme combinations of high power and low significance levels, but already for common combinations of desired power and significance. For example, the median sample size of a study employing a within-participants design would suffice to detect an effect of $d = .47$ with a power of .8 and at $\alpha = .05$. Obviously, the higher the desired power and the lower the selected α , the larger an effect must become to be detectable based on the sample sizes we found.

Furthermore, we aimed to determine the number of studies that report an a priori power analysis. The prevalence of a priori power analyses is distributed unequally over study categories (see Table 5): Whereas a majority of experimental studies report an a priori power analysis (65%), only a minority of quasi-experimental (41%) and correlational studies (26%) do.

Additionally, we aimed to explore whether studies that report an a priori power analysis employ larger sample sizes than studies that do not. Again, the results depend on the study categories (see Table 6): Experimental studies with an a priori power analysis have larger sample sizes than experimental studies without one, whereas quasi-experimental and correlational studies with an a priori power analysis have smaller sample sizes than respective studies without one.

4. Discussion

4.1. Discussion of results

The goal of the present paper was to investigate whether sample sizes in sport and exercise psychology have increased since the beginning of the current methodological discussion. Taken together, the present results suggest that sample sizes in sport and exercise psychology have substantially increased in the past decade. More precisely, both for experimental and for correlational studies, researchers have employed larger samples in the timespan between 2022 and 2023 than they did between 2009 and 2013. The data suggest that the reported increases are not part of random fluctuations, but that they instead represent a

Table 2

Median sample sizes per timespan (2009–2013 vs. 2022–2023), analyzed separately for experimental, quasi-experimental and correlational studies.

	2009–2013	2022–2023	Change (%)
Experimental (all)	40 (24; 72)	57.5 (34; 105.8)	44
Experimental (Within)	24 (14; 39)	37 (29; 81)	54
Experimental (Mixed)	47 (30; 85)	62 (34.2; 87.5)	32
Experimental (Between)	75 (44; 124)	132 (89.5; 161.5)	76
Quasi-Experimental	91 (45; 206)	68 (40; 146)	–25
Correlational	221 (124; 386)	295 (133; 485)	33

Note. Numbers in brackets are quartiles (i.e., Q25, Q75). Change (%) is the change in median sample sizes from 2009 to 2013 to 2022–2023.

¹ Again, it is important to keep in mind that some study designs are more frequent than others, both in the data from 2009 to 2023 and the data from 2022 to 2023. Thus, the data from experimental and correlational studies are more informative than the data from quasi-experimental studies, particularly so when broken down for the different years from 2009 to 2013.

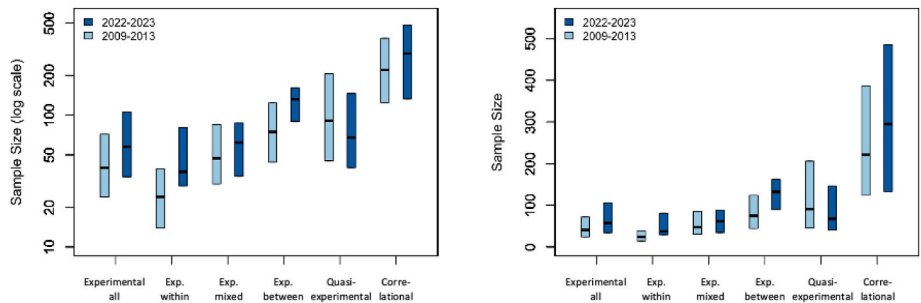


Fig. 1. Median sample sizes per timespan (2009-2013 vs. 2022-2023), analyzed separately for experimental, quasi-experimental and correlational studies. *Note.* The figure depicts median sample sizes, separately for different study categories (i.e., experimental, quasi-experimental and correlational studies). Due to the large differences between study categories, we present the data in two manners: The left-hand side of the figure depicts the data on a logarithmic scale, making it easier to see the differences between the timespans *within* one study category while downplaying the substantial differences between the study categories. The right-hand side of the figure depicts the data on a normal scale, thus making it harder to see the differences between the timespans, but highlighting the differences between the study categories.

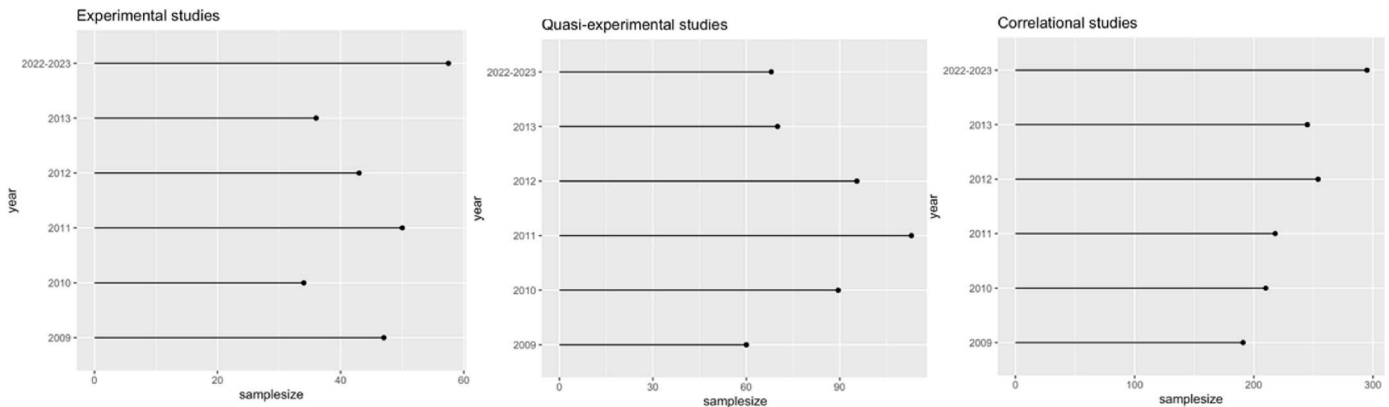


Fig. 2. Median sample sizes in 2022-2023 compared to the fluctuations between 2009 and 2013. *Note.* The figure depicts median sample sizes (on the x-axis) and years (on the y-axis), separately for different study categories (i.e., experimental, quasi-experimental and correlational studies).

Table 3
Statistical power to detect different effect sizes across study categories.

	$\alpha = .05$			$\alpha = .01$		
	$r = .1$ ($d = .20$)	$r = .2$ ($d = .41$)	$r = .3$ ($d = .63$)	$r = .1$ ($d = .20$)	$r = .2$ ($d = .41$)	$r = .3$ ($d = .63$)
Corr	.40 (.21; .60)	.93 (.64; .99)	.99 (.94; .99)	.19 (.08; .35)	.81 (.40; .97)	.99 (.83; .99)
Quasi-Exp	.12 (.09; .22)	.38 (.24; .69)	.73 (.49; .97)	.04 (.03; .08)	.18 (.09; .45)	.48 (.25; .88)
Exp-Within	.21 (.18; .43)	.68 (.57; .95)	.96 (.91; .99)	.08 (.06; .21)	.42 (.31; .85)	.86 (.73; .99)
Exp-Between	.21 (.16; .24)	.65 (.49; .74)	.95 (.84; .98)	.07 (.05; .09)	.40 (.25; .50)	.84 (.64; .92)
Exp-Mixed						
Within-Factor	.34 (.21; .46)	.89 (.65; .97)	.99 (.95; .99)	.15 (.07; .23)	.71 (.39; .88)	.99 (.83; .99)
Between-Factor	.15 (.10; .19)	.45 (.27; .59)	.80 (.55; .92)	.05 (.03; .07)	.22 (.11; .35)	.58 (.30; .78)
Interaction	.34 (.21; .46)	.89 (.65; .97)	.99 (.95; .99)	.15 (.07; .23)	.71 (.39; .88)	.99 (.83; .99)

Note. Power values were computed using the median sample sizes from Table 2 while assuming a range of population effect sizes ($r = .1 - .3$) for $\alpha = .05$ and $.01$. The reported power values assume a two-tailed test. Numbers in brackets denote the power of studies with sample sizes of Q25 and Q75, respectively.

systematic development. At the same time, the reported increases do not appear to be trivial, as they correspond to respective increases in power. For example, the average power of a within-participants experimental study increased for $d = .41$ and $\alpha = .05$ (all else equal) from .49 to .68. Only for quasi-experimental studies, sample sizes have decreased. Speculatively, this might be because researchers have limited control over their samples when conducting quasi-experimental studies and just have to “take who they can get”, e.g., already existing sports teams or classes. Since the amount of quasi-experimental studies was limited both in 2009-2013 and 2022-2023, the downward trend could thus be caused by such unsystematic factors. Studying publications from more than two years would have been helpful in order to distinguish systematic from

unsystematic factors. However, due to resource constraints, the present analysis is confined to two years.

One might argue that the estimates of power and MDES presented in this paper are superfluous: As power is a function of sample sizes, effect sizes and a priori significance levels, these calculations do not add any novel information to a simple reporting of sample sizes. However, they serve to *illustrate* the influence of sample sizes, both on power and on MDES. We assume that these illustrations might be helpful for readers in order to try and gauge the impact of sample sizes on metrics that they care about, such as power.

Additionally, our results suggest that still only 42% of studies report an a priori power analysis, and that the prevalence of a priori power

Table 4
Minimum detectable effect sizes (MDES) across study categories.

	Power = .8		Power = .95	
	$\alpha = .05$	$\alpha = .01$	$\alpha = .05$	$\alpha = .01$
Corr (r)	.16 (.24; .13)	.2 (.29; .15)	.21 (.31; .16)	.24 (.35; .19)
Quasi-Exp (d)	.69 (.91; .47)	.85 (1.13; .57)	.89 (1.17; .6)	1.05 (1.4; .71)
Exp-Within (d)	.47 (.54; .32)	.59 (.68; .39)	.61 (.69; .41)	.73 (.83; .48)
Exp-Between (d)	.49 (.6; .44)	.6 (.74; .54)	.63 (.77; .57)	.74 (.91; .67)
Mixed				
Within-factor (f)	.18 (.25; .15)	.22 (.31; .19)	.23 (.32; .19)	.28 (.38; .23)
Between-factor (f)	.31 (.43; .26)	.39 (.53; .32)	.4 (.55; .34)	.48 (.66; .4)
Interaction-factor (f)	.18 (.25; .15)	.22 (.31; .19)	.23 (.32; .19)	.28 (.38; .23)

Note. MDES were computed using the median sample sizes from Table 2 while assuming two different levels of power (.8 and .95) for $\alpha = .05$ and .01. Numbers in brackets denote the MDES detectable in studies with sample sizes of Q25 and Q75, respectively.

Table 5
Number of studies with and without an a priori power analysis, displayed separately for experimental, quasi-experimental and correlational studies.

	Without a priori power analysis	With a priori power analysis
Experimental (all)	42 (35)	78 (65)
Quasi-Experimental	22 (59)	15 (41)
Correlational	125 (74)	44 (26)
Total	189 (58)	137 (42)

Note. Numbers in brackets indicate percentages (%).

Table 6
Median sample sizes for studies with and without an a priori power analysis, depicted separately for experimental, quasi-experimental and correlational studies.

	Without a priori power analysis	With a priori power analysis
Experimental (all)	50.5	61.5
Experimental (Within)	37	39.5
Experimental (Mixed)	50	70
Experimental (Between)	184.5	117
Quasi-Experimental	91	42
Correlational	308	230.5

analyses is distributed unequally over study categories. Given the amount of literature and discussions surrounding the power issue, it seems unlikely that the reported lack of power analyses is due to its underestimation by researchers. Rather, we assume that more practical training is needed on how to perform power analyses for specific designs and in absence of strong prior information on effect sizes. Luckily, the field seems to be moving in this direction (e.g., Lakens & Caldwell, 2021; Sommet et al., 2023).

It seems important to note that power analyses do not serve the goal of “making samples larger”, this should only be the case when a field is systematically underpowered (Miller & Ulrich, 2016). Instead, power analyses are one means of planning and justifying a sample size (Lakens, 2022). Thus, researchers could also learn from an a priori power analysis that they need fewer participants than they previously thought. This might be the case for researchers investigating large effects in correlational studies or in applied studies. With the benefit of hindsight, the

data reported above seem to support this idea: Correlational studies with power analyses employed smaller sample sizes than correlational studies without power analyses, potentially because correlational studies were rather adequately powered in the first place. However, there are some crucial caveats to this reasoning: First, effect sizes are probably on average rather small in psychological research (Richard et al., 2003; Sommet et al., 2023), so most research will probably not be looking for effects on the right-hand side of the table. Second, we calculated power for detecting single correlations in the whole sample. Once researchers start looking for more complicated effects (e.g., interactions, or regression effects based on intercorrelated predictors) or into subsamples (e.g., based on gender or occupation), power will be lower again.

Finally, and not directly related to the goal of the present paper, we were surprised to learn that only 8% of the papers we analyzed were multi-study papers. In other words, 85% of studies appeared in single-study papers. In light of the current discussion regarding the ever-growing number of papers that need to be reviewed (Hanson et al., 2024), this number seems worrisome. Publishing more multi-study papers might at least partly contribute to reducing the strain on the publication system. At the same time, it seems important to acknowledge that publishing multi-study papers may be more difficult for some groups of researchers (e.g., PhD students who have to finish their project in time) than for others (e.g., tenured researchers). Furthermore, it may be easier in some fields (e.g., experimental laboratory research) than in others (e.g., field research in high-performance sports). Therefore, when trying to increase the proportion of multi-study papers one needs to be careful not to create incentives that are good-willed, but ultimately problematic for the field (e.g., discouraging researchers from conducting studies in high-performance sports).

4.2. Limitations

The present research has some limitations. First, and most importantly, the reported changes cannot be interpreted causally, as the data are observational (see Hardwicke et al., 2020, on the challenges of evaluating research reforms without proper evaluation designs). Thus, tempting as it may be, we cannot say that sample sizes in sport and exercise psychology have increased *because* of the methodological discussions in the past decade. However, we can say that they *did* increase. Furthermore, it would have been preferable to not only compare 2022-2023 to 2009-2013, but instead to analyze all years between 2014 and 2023. This approach might have allowed more insights into causal developments, e.g., if we had seen a sudden increase in sample sizes in a certain year, similar to a regression discontinuity effect (Cattaneo & Titiunik, 2022). Unfortunately, due to resource constraints, this approach was not feasible, particularly so as all studies were coded by two authors in order to ensure comparability with the data from 2009 to 2013.

Second, all comparisons regarding studies' power between 2009-2013 and 2022-2023 depend on the assumption that, on average, the true sizes of the effects that authors were investigating did not change. If, say, researchers in 2022-2023 had investigated smaller effects than in 2009-2013, then power might not have increased due to larger samples, but instead it might have stayed constant or even decreased. However, there is no reason to assume that researchers in the field of sport and exercise psychology investigated smaller effects in the respective period. Likewise, comparisons regarding power depend on the assumption that alpha levels did not change between 2009-2013 and 2022-2023. If researchers had lowered their alphas (as suggested by Benjamin et al., 2018), then sample sizes cannot be compared without further corrections for the respective timespan. Unfortunately, controlling for alpha levels was not possible, as the respective information is not available for the studies from 2009 to 2013. Future studies might want to take alpha levels into account.

Third, just as in Furley and Schweizer (2016), all estimates of power are based on simplifications and assumptions. For example, we estimate

power for certain designs (e.g., within-participants designs with two within-measurements) and we assume certain effect sizes (see the respective online supplement [<https://doi.org/10.23668/psycharchives.22334>] for the assumptions underlying estimates of power). To the extent that studies deviate from these assumptions, we either over- or underestimate their true power. For example, when researchers calculate correlation coefficients for subsamples, these analyses will have lower power than the analyses for the whole sample. We chose our approach to be as benevolent as possible, thus we should rather over- than underestimate studies' true power. Again, differences between 2009–2013 and 2022–2023 will affect the comparison between the two timespans. Furthermore, researchers sometimes conduct multiple significance tests regarding one hypothesis on the same sample, without correcting for the inflation of Type-1 errors caused by multiple testing. The more widespread this practice in a field, the more false positive results should emerge in this field. However, in the present analyses, we did not control for multiple testing or for the presence of proper statistical corrections. When planning a study or conducting a priori power analyses, it seems sensible to take multiple testing into account and to employ proper correction procedures.

One major difference we observed between the two measurement times is the increase in multilevel models over time. In the data from 2022 to 2023, there are substantially more multilevel models than in the data from 2009 to 2013, probably reflecting a development in research methodology in the field. Power calculations for multilevel models are more complex than for experiments or correlational studies (e.g., Murayama et al., 2022). In line with this observation, several authors noted that they found it not feasible to conduct an a priori power analysis for their multilevel models. Thus, developing guidelines and software for power analyses for multilevel models seems to be a worthwhile endeavor (see Murayama et al., 2022, for a practical method and software for power analyses for simple multilevel models). For the present study, we treated multilevel models as correlational studies in order to enable comparisons between the timespans. Due to their fundamentally correlational nature, treating multilevel models as correlational studies for the present purposes seemed a reasonable approximation from a statistical perspective. Indeed, Murayama et al. (2022) illustrate mathematically that it can be appropriate to use the correlation coefficient r for power calculations for simple multilevel models. For more complex models, particularly models with inter-correlated predictors, this approach might underestimate the sample sizes required to obtain a specific level of power.

Finally, one might question whether some studies had small sample sizes because researchers sampled entire populations (Lakens, 2022). However, we did not find evidence for this strategy, particularly so as all studies reported significance tests, which is not appropriate when investigating whole populations. Still, one needs to be careful not to question a study's power when the respective study does not rely on a sample but on an entire population (Lakens, 2022).

In order to avoid misunderstandings, we would like to emphasize that increasing sample sizes is not the only way to increase power. Among other factors, power also depends on studies' designs (e.g., all else equal, within-participants designs have higher power than between-participants designs) and the reliabilities of the measured variables (the more reliable the measurement, the higher the power). Increasing the reliabilities of measured variables has substantial effects on power (see Sommet et al., 2023, for an illustration). Furthermore, power may depend on the exact nature of the investigated effect (e.g., the shape of an interaction needs to be considered when planning sample sizes of studies researching interactions [Schweizer & Köppel, 2025; Sommet et al., 2023]) and on the relationships between variables (e.g., the intercorrelations between predictors in a regression study need to be considered when planning sample sizes [Maxwell, 2000]).

5. Conclusions and outlook

While it does not allow for causal interpretations, the present study suggests that sample sizes in sport and exercise psychology (with the exception of quasi-experimental studies) increased substantially in the past decade. This observation is not in line with the frequently expressed idea that researchers do not change their research practices (e.g., Hardwicke et al., 2022; Tiller & Ekkekakis, 2023). Rather, it is in line with the more optimistic perspective that researchers do change their research practices, if they are both motivated *and* capable to do so (Nelson et al., 2018). This particular change in research practices may have led to an increase in power for experiments and correlational studies in sport and exercise psychology, which in turn should have reduced both false-positive and false-negative results. Although the present results are promising, there are still some reasons to be concerned. First and foremost, researchers still do not report an a priori power analysis for a majority of studies. Even in the category of studies with the highest proportion of a priori power analyses (i.e., experiments), one third of studies were published without mentioning one. Maybe journals will really need to change their editorial policies in order to remedy this shortcoming (Brysbaert, 2019; Nosek et al., 2015).

For the future of sport and exercise psychology, we hope that researchers will continue to care about the quality of their research, and that progress in education and practical applications will make it easier to conduct meaningful power analyses for all kinds of designs. In this case, research in sport and exercise psychology may not need to be reborn one day, as suggested by Nelson et al. for the field of psychology as a whole (2018), but instead it may continue to evolve and prosper.

Funding

The authors received no funding.

CRediT authorship contribution statement

Tadeus Pindl: Data curation, Formal analysis, Writing – original draft. **Geoffrey Schweizer:** Conceptualization, Data curation, Supervision, Writing – original draft. **Philip Furley:** Conceptualization, Writing – original draft.

Declaration of competing interest

The authors have no conflicts of interest to declare (financial or otherwise). The authors received no funding.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.23668/psycharchives.22334>.

Data availability

The raw data and the analysis script are available at PsychArchives (link provided in the manuscript).

References

- Benjamin, D. J., Berger, J. O., Johannesson, M., Nosek, B. A., Wagenmakers, E.-J., Berk, R., Bollen, K. A., Brembs, B., Brown, L., Camerer, C., Cesarini, D., Chambers, C. D., Clyde, M., Cook, T. D., De Boeck, P., Dienes, Z., Dreber, A., Easwaran, K., Efferson, C., ... Johnson, V. E. (2018). Redefine statistical significance. *Nature Human Behaviour*, 2, 6–10. <https://doi.org/10.1038/s41562-017-0189-z>
- Bloom, H. S. (1995). Minimum detectable effects: A simple way to report the statistical power of experimental designs. *Evaluation Review*, 19, 547–556. <https://doi.org/10.1177/0193841x9501900504>
- Brysbaert, M. (2019). How many participants do we have to include in properly powered experiments? A tutorial of power analysis with reference tables. *Journal of Cognition*, 2(1). <https://doi.org/10.5334/joc.72>

- Button, K. S., Ioannidis, J. P. A., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S. J., & Munafò, M. R. (2013). Power failure: Why small sample sizes undermine the reliability of neuroscience. *Nature Reviews Neuroscience*, 14, 365–376. <https://doi.org/10.1038/nrn3475>
- Cattaneo, M. D., & Titiunik, R. (2022). Regression discontinuity designs. *Annual Review of Economics*, 14, 821–851. <https://doi.org/10.1146/annurev-economics-051520-021409>
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G* power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39, 175–191. <https://doi.org/10.3758/BF03193146>
- Friley, R. C., & Vazire, S. (2014). The N-Pact factor: Evaluating the quality of empirical journals with respect to sample size and statistical power. *PLoS One*, 9, Article e109019. <https://doi.org/10.1371/journal.pone.0109019>
- Hanson, M. A., Barreiro, P. G., Crosetto, P., & Brockington, D. (2024). The strain on scientific publishing. *Quantitative Science Studies*, 5(4), 823–843. <https://doi.org/10.1162/qss.a.00327>
- Hardwicke, T. E., Serghiou, S., Janiaud, P., Danchev, V., Crüwell, S., Goodman, S. N., & Ioannidis, J. P. A. (2020). Calibrating the scientific ecosystem through meta-research. *Annual Review of Statistics and its Application*, 7, 11–37. <https://doi.org/10.1146/annurev-statistics-031219-041104>
- Hardwicke, T. E., Thibault, R. T., Kosie, J. E., Wallach, J. D., Kidwell, M. C., & Ioannidis, J. P. A. (2022). Estimating the prevalence of transparency and reproducibility-related research practices in psychology (2014–2017). *Perspectives on Psychological Science*, 17(1), 239–251. <https://doi.org/10.1177/1745691620979806>
- Hardwicke, T. E., & Wagenmakers, E.-J. (2023). Reducing bias, increasing transparency and calibrating confidence with preregistration. *Nature Human Behaviour*, 7, 15–26. <https://doi.org/10.1038/s41562-022-01497-2>
- Kossmeier, M., Vilsmeier, J., Dittrich, R., Fritz, T., Kolmanz, C., Plessen, C. Y., Slowik, A., Tran, U. S., & Voracek, M. (2019). Long-term trends (1980–2017) in the *n*-pact factor of journals in personality psychology and individual differences research. *Zeitschrift für Psychologie*, 227(4), 293–302. <https://doi.org/10.1027/2151-2604/a000384>
- Lakens, D. (2014). Observed power, and what to do if your editor asks for post-hoc power analyses. *The 20% Statistician*. <https://daniellakens.blogspot.com/2014/12/observe-d-power-and-what-to-do-if-your.html>
- Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, 8. <https://doi.org/10.1525/collabra.33267>
- Lakens, D., Scheel, A. M., & Isager, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, 1(2), 259–269. <https://doi.org/10.1177/2515245918770963>
- Lakens, D., & Caldwell, A. R. (2021). Simulation-based power analysis for factorial analysis of variance designs. *Advances in Methods and Practices in Psychological Science*, 4(1). <https://doi.org/10.1177/2515245920951503>
- Lakens, D., & Evers, E. R. K. (2014). Sailing from the seas of chaos into the corridor of stability: Practical recommendations to increase the informational value of studies. *Perspectives on Psychological Science*, 9(3), 278–292. <https://doi.org/10.1177/1745691614528520>
- Maxwell, S. E. (2000). Sample size and multiple regression analysis. *Psychological Methods*, 5(4), 434–458. <https://doi.org/10.1037/1082-989X.5.4.434>
- Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2022). Replication concerns in sports and exercise science: A narrative review of selected methodological issues in the field. *Royal Society Open Science*, 9, Article 220946. <https://doi.org/10.1098/rsos.220946>
- Milkman, Katherine L., Gromet, Dena, Ho, Hung, Kay, Joseph S., Lee, Timothy W., Pandiloski, Pepi, Park, Yeji, Rai, Aneesh, Bazerman, Max, Beshears, John, Bonacorsi, Lauri, Camerer, Colin, Chang, Edward, Chapman, Gretchen, Cialdini, Robert, Dai, Hengchen, Eskreis-Winkler, Lauren, Fishbach, Ayelet, Gross, James J., ... Duckworth, Angela L. (2021). Megastudies improve the impact of applied behavioural science. *Nature*, 600(7889), 478–483. <https://doi.org/10.1038/s41586-021-04128-4>. (Accessed 8 December 2021)
- Miller, J., & Ulrich, R. (2016). Optimizing research payoff. *Perspectives on Psychological Science*, 11(5), 664–691. <https://doi.org/10.1177/1745691616649170>
- Munafò, M. R., Nosek, B. A., Bishop, D. V. M., Button, K. S., Chambers, C. D., Percie du Sert, N., Simonsohn, U., Wagenmakers, E.-J., Ware, J. J., & Ioannidis, J. P. A. (2017). A manifesto for reproducible science. *Nature Human Behaviour*, 1, 21. <https://doi.org/10.1038/s41562-016-0021>
- Murayama, K., Usami, S., & Sakaki, M. (2022). Summary-statistics-based power analysis: A new and practical method to determine sample size for mixed-effects modeling. *Psychological Methods*, 27(6), 1014–1038. <https://doi.org/10.1037/met0000330>
- Murphy, J., Caldwell, A. R., Mesquida, C., Ladell, A. J. M., Encarnación-Martínez, A., Tual, A., Denys, A., Cameron, B., Van Hooren, B., Parr, B., DeLucia, B., Mason, B. R. J., Clark, B., Egan, B., Brown, C., Ade, C., Sforza, C., Taber, C. B., Kirk, C., McCrum, C., ... Warne, J. P. (2025). Estimating the replicability of sports and exercise science research. *Sports Medicine*, 55, 2659–2679. <https://doi.org/10.1007/s40279-025-02201-w>
- Nelson, L. D., Simmons, J., & Simonsohn, U. (2018). Psychology's renaissance. *Annual Review of Psychology*, 69, 511–534. <https://doi.org/10.1146/annurevpsych-122216-011836>
- Nosek, B. A., Alter, G., Banks, G. C., Borsboom, D., Bowman, S. D., Breckler, S. J., ... Yarkoni, T. (2015). Promoting an open research culture. Author guidelines for journals could help to promote transparency, openness, and reproducibility. *Science*, 348, 1422–1425. <https://doi.org/10.1126/science.aab2374>
- R Core Team. (2023). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>
- Reardon, K. W., Smack, A. J., Herzhoff, K., & Tackett, J. L. (2019). An *N*-pact factor for clinical psychological research. *Journal of Abnormal Psychology*, 128(6), 493–499. <https://doi.org/10.1037/abn0000435>
- Richard, F. D., Bond Jr, C. F., & Stokes-Zoota, J. J. (2003). One hundred years of social psychology quantitatively described. *Review of General Psychology*, 7, 331–363. <https://doi.org/10.1037/1089-2680.7.4.331>
- Schäfer, Thomas, & Schwarz, Marcus A. (2019). The meaningfulness of effect sizes in psychological research: Differences between sub-disciplines and the impact of potential biases. *Front. Psychol.*, 10, Article 813. <https://doi.org/10.3389/fpsyg.2019.00813>. <https://www.frontiersin.org/article/10.3389/fpsyg.2019.00813/full>. (Accessed 11 April 2019)
- Schimmack, U. (2021). The validation crisis in psychology. *Meta-Psychology*, 5, 1645. <https://doi.org/10.15626/MP.2019.1645>. MP.2019.
- Schweizer, G., Furlley, P., Rost, N., & Barth, K. (2020). Reliable measurement in sport psychology: The case of performance outcome measures. *Psychology of Sport and Exercise*, 48, Article 101663. <https://doi.org/10.1016/j.psychsport.2020.101663>
- Schweizer, G., & Furlley, P. (2016). Reproducible research in sport and exercise psychology: The role of sample sizes. *Psychology of Sport and Exercise*, 23, 114–122. <https://doi.org/10.1016/j.psychsport.2015.11.005>
- Schweizer, G., & Köppel, M. (2025). In search of the lost interaction: A theoretical and methodological framework for researching interactions. *Europe's Journal of Psychology*, 21, 249–262. <https://doi.org/10.5964/ejop.14957>
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22, 1359–1366. <https://doi.org/10.1016/10.1177/0956797611417632>
- Sommet, N., Weissman, D. L., Cheutin, N., & Elliot, A. J. (2023). How many participants do I need to test an interaction? Conducting an appropriate power analysis and achieving sufficient power to detect an interaction. *Advances in Methods and Practices in Psychological Science*, 6. <https://doi.org/10.1177/25152459231178728>
- Tiller, N. B., & Ekkekakis, P. (2023). Overcoming the “ostrich effect”: A narrative review on the incentives and consequences of questionable research practices in kinesiology. *Kinesiology Review*, 12(3), 201–216. <https://doi.org/10.1123/kr.2022-0039>
- Uehata, A., & Rein, R. (2025). Statistical power analysis in exercise science. *Journal of Sports Sciences*, 1–17. <https://doi.org/10.1080/02640414.2025.2571841>
- Wagenmakers, E.-J., Marsman, M., Jamil, T., Ly, A., Verhagen, A. J., Love, J., Selker, R., Gronau, Q. F., Smíra, M., Epskamp, S., Matzke, D., Rouder, J. N., & Morey, R. D. (2018). Bayesian inference for psychology. Part I: Theoretical advantages and practical ramifications. *Psychonomic Bulletin & Review*, 25, 35–57. <https://doi.org/10.3758/s13423-017-1343-3>
- Westfall, J., & Yarkoni, T. (2016). Statistically controlling for confounding constructs is harder than you think. *PLoS One*, 11, Article e0152719. <https://doi.org/10.1371/journal.pone.0152719>
- Wilson, B. M., & Wixted, J. T. (2018). The prior odds of testing a true effect in cognitive and social psychology. *Advances in Methods and Practices in Psychological Science*, 1, 186–197. <https://doi.org/10.1177/2515245918767122>
- Zhang, Z., & Mai, Y. (2023). *WebPower: Basic and advanced statistical power analysis*. <https://CRAN.R-project.org/package=WebPower>.